Cisco Packet Tracer – VLANs and Static Routing

1. Create Two VLANs: Sales and Support

Open Cisco Packet Tracer and add one Cisco 2960 switch.

1. Click the switch and open the CLI tab.
2. Enter the following commands to create the VLANs:

```
enable
configure terminal

vlan 10
name Sales
exit

vlan 20
name Support
exit
```

VLAN 10 is named Sales, and VLAN 20 is named Support.

2. Assign Switch Ports and Verify VLAN Membership

Assign three ports to Sales and two ports to Support.

```
configure terminal

interface range fastEthernet 0/1-3
switchport mode access
switchport access vlan 10
exit

interface range fastEthernet 0/4-5
switchport mode access
switchport access vlan 20
exit
```

end

show vlan brief

Expected VLAN membership:

VLAN ID	VLAN Name	Assigned Ports
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

The show vlan brief command verifies that the ports are assigned to the correct VLANs.

3. Connect Two Routers and Configure Static Routing

Network diagram

Router 1 (R1)

LAN: 192.168.1.1/24

Link: 10.0.0.1/30

Ethernet link: 10.0.0.0/30

Router 2 (R2)

LAN: 192.168.2.1/24

Link: 10.0.0.2/30

Use Ethernet interfaces for this example. Connect the routers using a suitable Ethernet cable, and ensure the interface names match your router model.

Router 1 configuration

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.1 255.255.255.252

no shutdown

exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end

Router 2 configuration

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1

end

Verify the routes

On each router, run:

show ip interface brief

show ip route

The static route on R1 tells it to forward packets destined for 192.168.2.0/24 to R2 at 10.0.0.2. R2 has the reverse route for 192.168.1.0/24.

4. Test Ping Between Sales and Support VLANs

Suppose:

- Sales PC: 192.168.10.10/24, connected to Fa0/1 in VLAN 10.
- Support PC: 192.168.20.10/24, connected to Fa0/4 in VLAN 20.

From the Sales PC, open Desktop → Command Prompt and run:

ping 192.168.20.10

Expected result: The ping fails if there is no Layer 3 routing between VLAN 10 and VLAN 20.

Why? Each VLAN is a separate Layer 2 broadcast domain. A regular Layer 2 switch cannot route IP packets between different VLANs by itself.

How to enable inter-VLAN communication

You can use either of these methods:

- Router-on-a-stick: Connect a router to the switch using a trunk link, configure router subinterfaces for VLANs 10 and 20, and assign each subinterface an IP address in its VLAN's subnet.
- Layer 3 switch: Create switched virtual interfaces (SVIs) for VLANs 10 and 20, assign gateway IP addresses, and enable IP routing.

For router-on-a-stick, the switch port connected to the router must be configured as a trunk, and the router subinterfaces must use the matching VLAN IDs. PCs must also have the correct IP addresses, subnet masks and default gateways.

Conclusion: VLANs separate Sales and Support traffic. Inter-VLAN routing through a router or Layer 3 switch allows the two groups to communicate when required.